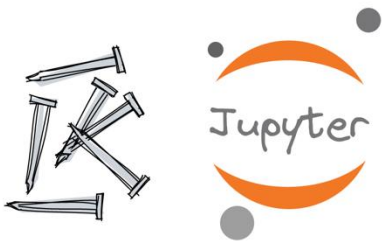
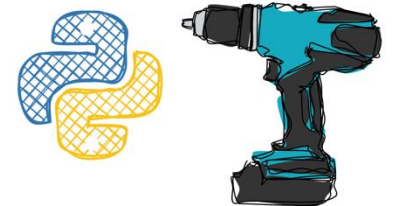


Building Your Own Custom ChatGPT with LoRA & RAG



Kader Mohideen





Idea

Create Digital Twin
of Imp Ex-Employee
in an Company

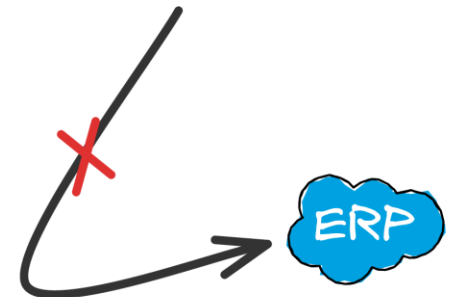
Problem Statement

Incident
History

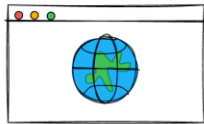
Projects History

Customer Discussion

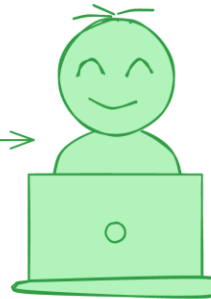
Renewals



Priya
Ex-Employee
NorthWIND SaaS
VP - Customer Relations

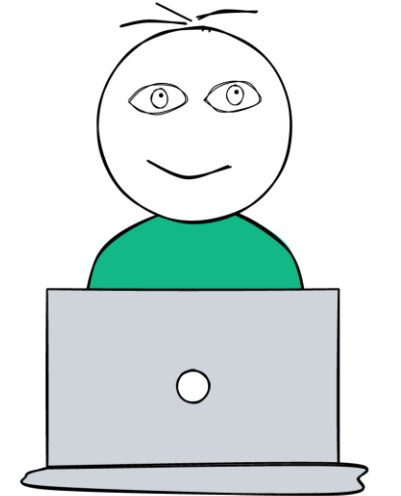


LLM



Digital Twin
of Priya

1. Understand History
2. Learn about Her Role & Task More
3. Respond to Customer Request Better



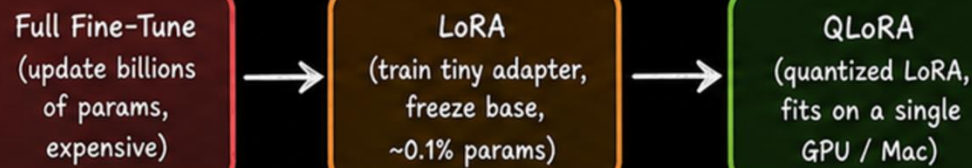
Arjun
Successor for Priya

Evolution of Models

1 ARCHITECTURE EVOLUTION



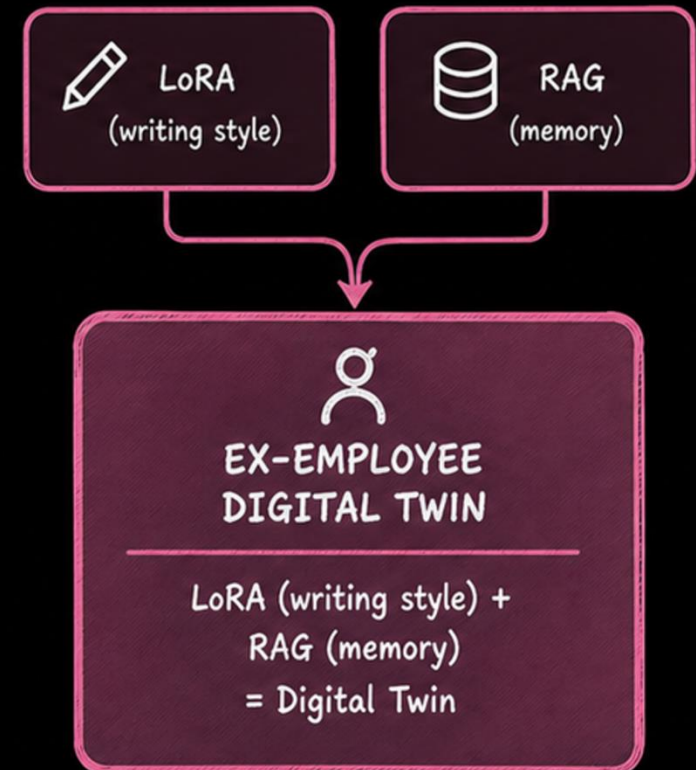
2 FINE-TUNING EVOLUTION



3 OUTPUTS (What We Get)



4 OUR PROJECT

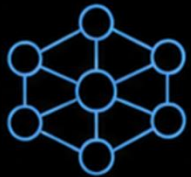


How LoRA Works

LoRA lets us fine-tune a big model by learning small “adapter” updates instead of changing all weights.

① Pretrained Model

We start with a pretrained model (e.g., Qwen).



Weights

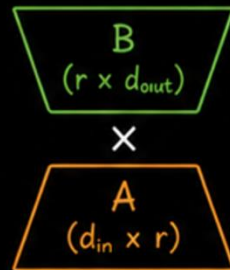
W (Frozen)

These weights are kept frozen.



② Add LoRA Adapters

We add small trainable adapters to certain layers.



B and A are small trainable matrices.



③ Low-Rank Update

The adapters create a low-rank update:

$$\Delta W = B \times A$$

This update is added to the frozen weights.

Effective Weights

$$W + \Delta W$$



④ Training

During training, only A and B are updated.



Frozen



Trainable



⑤ Inference

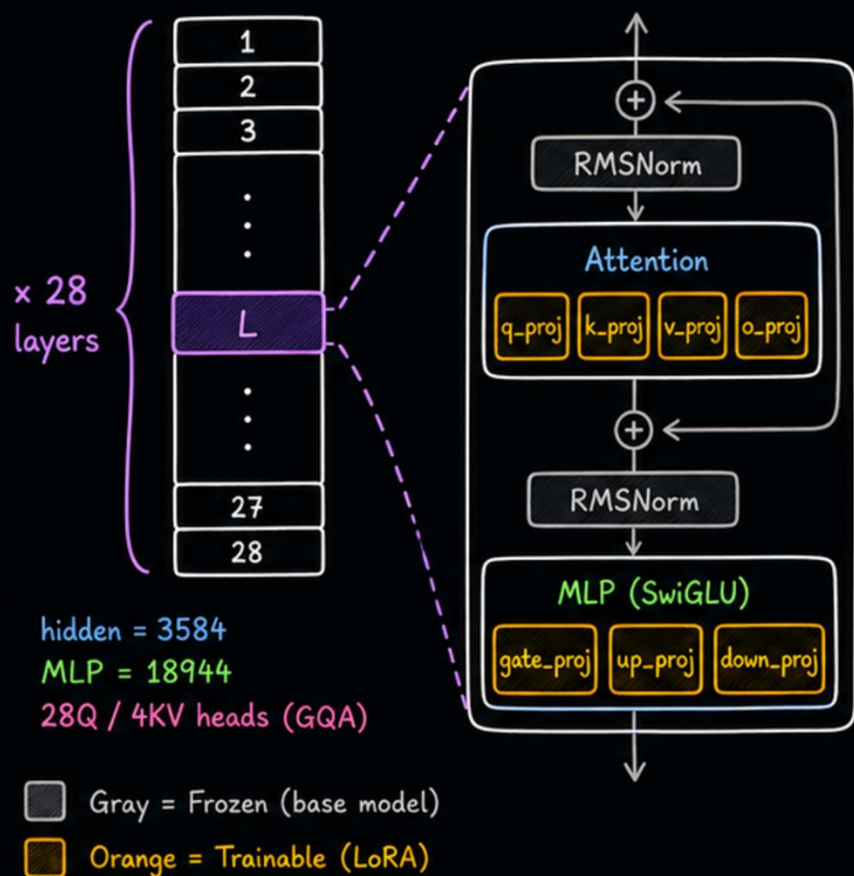
At inference, the model uses the updated weights:

$$W + \Delta W$$

No extra cost at inference time.



1 LoRA in Qwen2.5-7B



2 LoRA target matrices (drawn to scale)

Attention projections



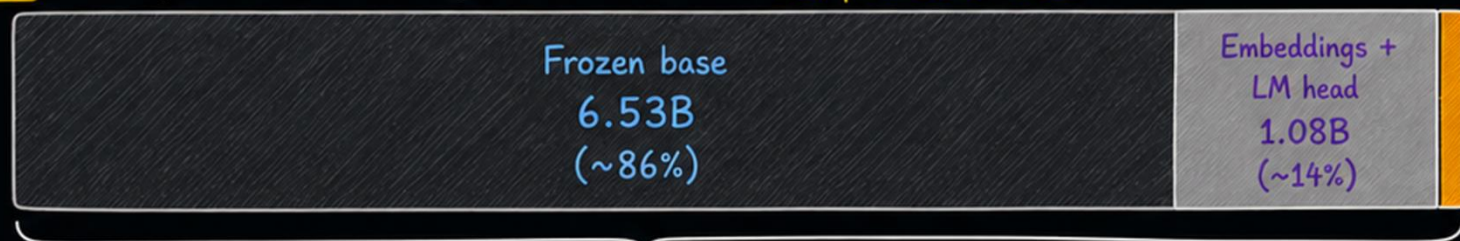
MLP (SwiGLU) projections



LoRA uses rank $r = 8$ for all 7 matrices ($B \in \mathbb{R}^{\text{out} \times 8}$, $A \in \mathbb{R}^{8 \times \text{in}}$)

3 LoRA parameter accounting (Qwen2.5-7B)

Total parameters: 7.61B



per layer: 233M frozen + 720,896 trainable $\cdot$ 196 adapter pairs total

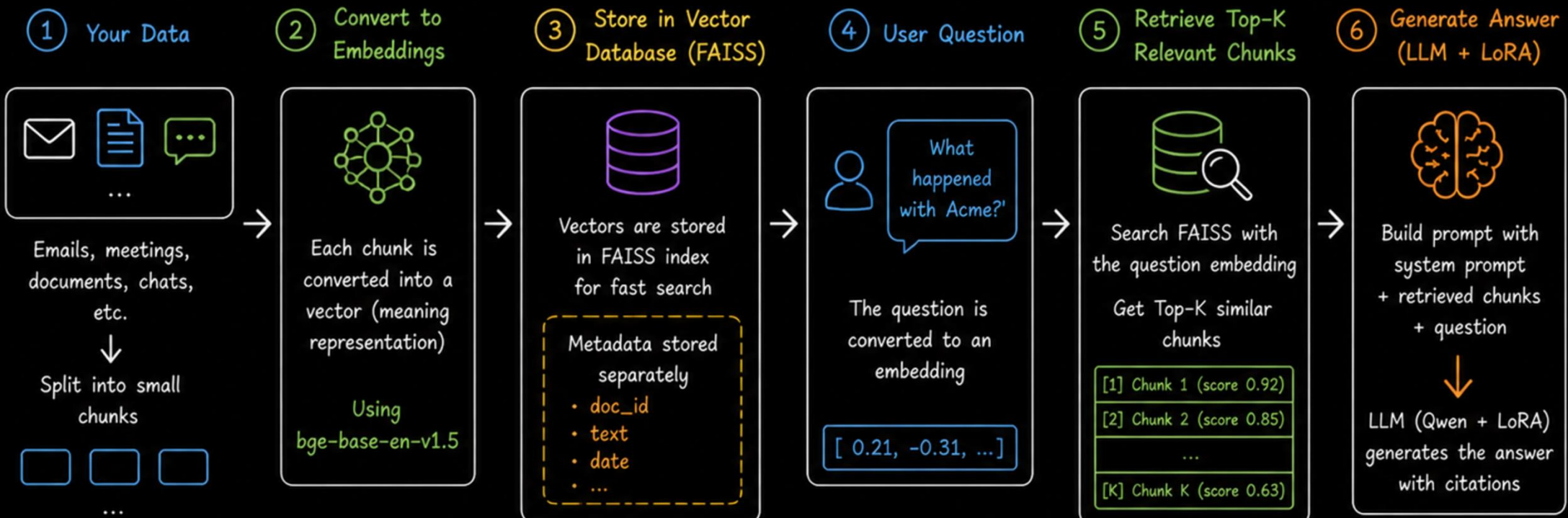
Gray = Frozen (not trained)
Orange = Trainable (LoRA)

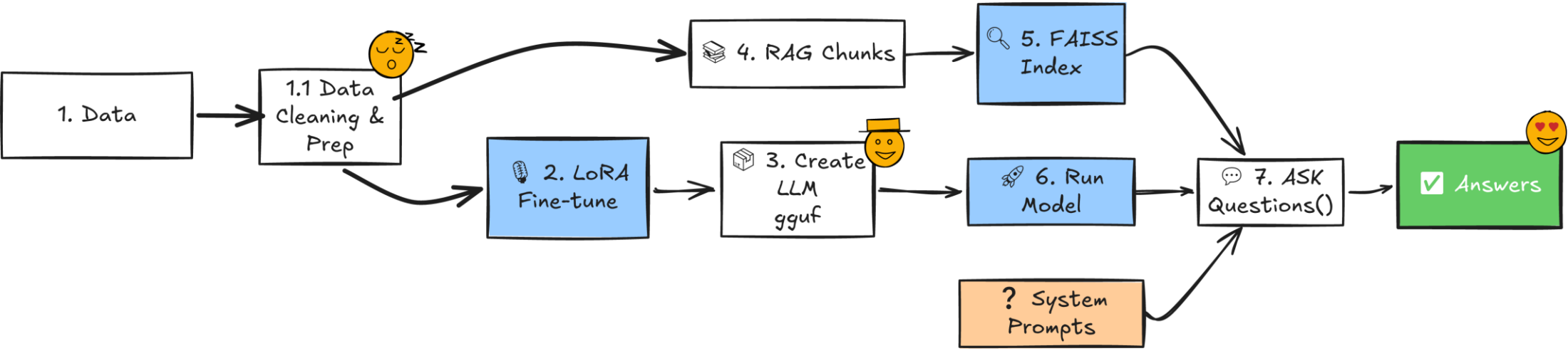
0.31%
trains
(20.19M of 6.55B)

LoRA
20.19M
 $\leftarrow$ trains

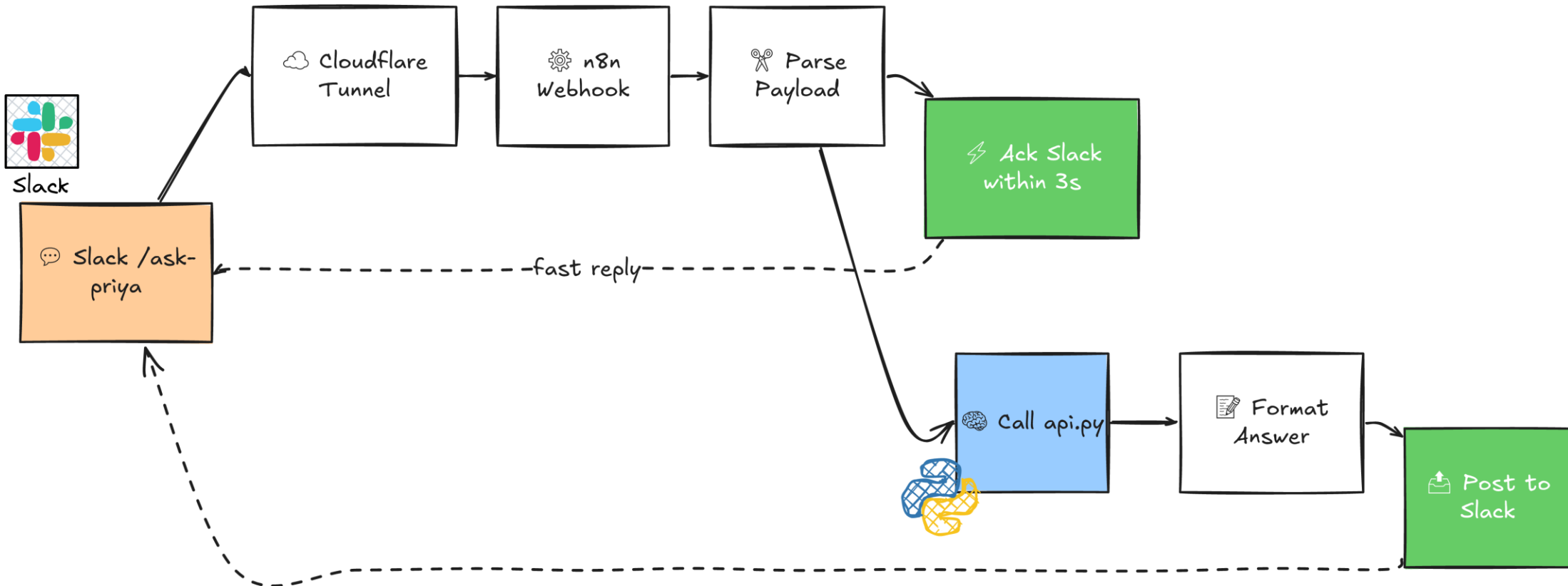
How RAG Works

Retrieve relevant information from your data and give it to the LLM to generate accurate answers.





Slack Integration



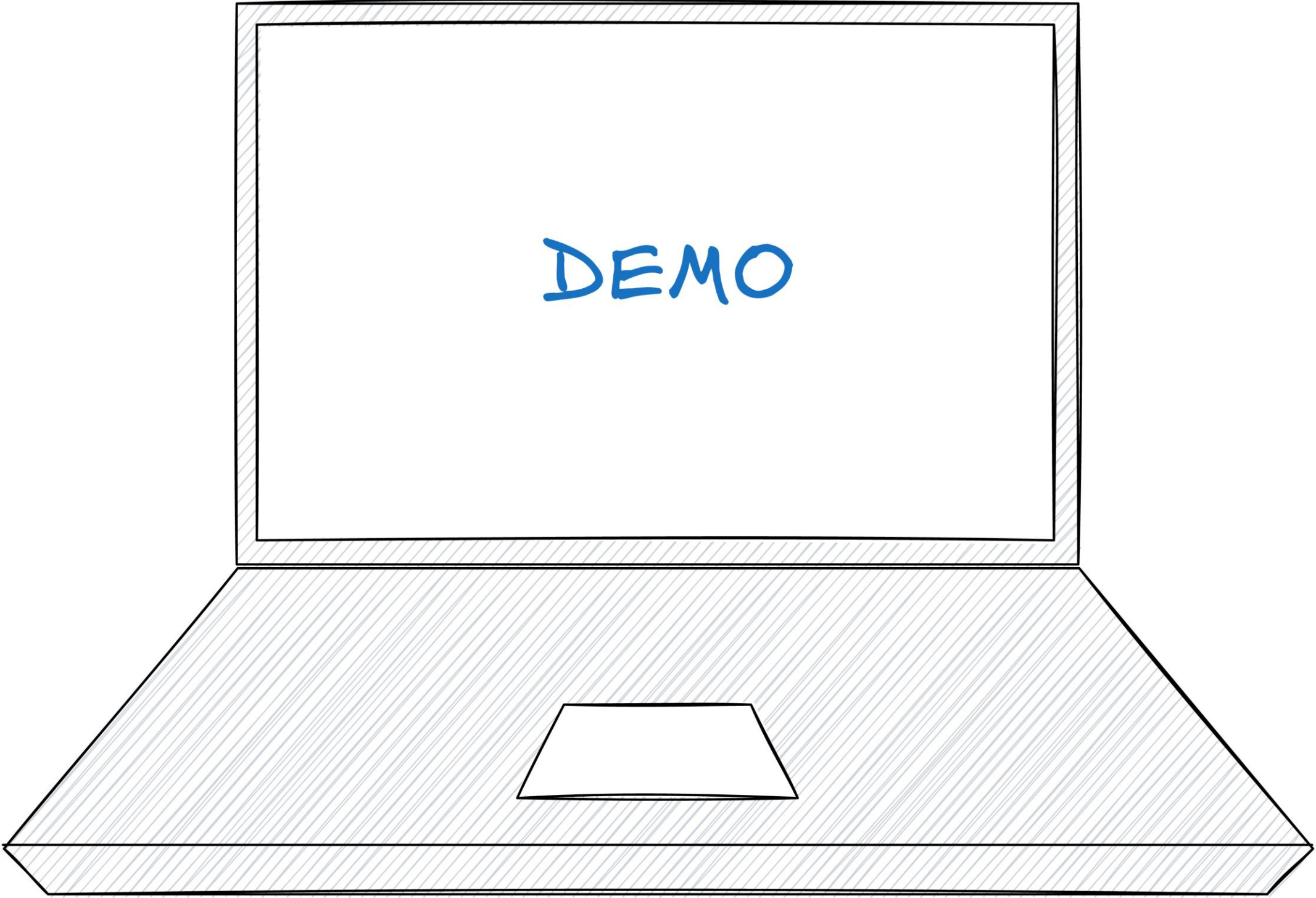
Data Prepation

Total Corpus

LAYER	FILES	DOCS	SIZE
1. Personas (priya, rohan)	2	–	10.1 KB
2. Cast (internal)	1	~25	3.1 KB
3. Accounts + Projects	2	40+14	22.5 KB
4. Corpus – priya/*.jsonl	19	16,779	11.7 MB
4. Corpus – rohan/*.jsonl	10	2,199	1.5 MB
= Total semantic docs	29	18,978	13.2 MB

Sample

```
{
  "doc_id": "email-acme-checkin-2024q1",
  "doc_type": "email",
  "thread_id": "th-acme-checkin-2024q1",
  "thread_position": 1,
  "date": "2024-03-12T17:14:00-05:00",
  "from": {"name": "Priya Sharma", "email": "priya.sharma@northwind-saas.com"},
  "to": [{"name": "Sarah Lin", "email": "sarah.lin@acmecorp.com"}],
  "subject": "Acme Corp – quick March check-in",
  "body": "Hi Sarah,\n\nQuick quarterly check-in. From our side things have been steady – usage holding trending up...\n\n• walk through the new audit-log feature shipping next month\n• align on adoption goals for next quarter\n\nWould 30 min late March work to do a proper review? Calendly: cal.com/priyasharma.\n\nBest,\nPriya",
  "tags": ["routine", "quarterly-checkin"],
  "related_account": "acme"
}
```



A hand-drawn illustration of a laptop. The screen is white and displays the word "DEMO" in blue, handwritten-style capital letters. The laptop's body is light blue with diagonal hatching. The keyboard area is a white trapezoid centered on the base. The entire device is outlined in black.

DEMO

Use case (Projects Priya Handled)

```
_ = ask('List every customer in your knowledge base and their renewal date, sorted by ARR descending.')
```

✓ 11.0s

ANSWER

=====

Here are the customers in my knowledge base and their renewal dates:

1. Acme Corp – 2024-12-31 [Source 5]
2. BuyNLarge Retail – 2024-12-31 [Sources 2, 4, 7]
3. ENCOM Software – 2025-12-31 [Source 8]
4. Abstergo Industries – 2026-12-31 [Source 3]
5. Wonka Confections – 2025-09-30 [Sources 1, 6]

Sorted by ARR descending (estimated):

1. BuyNLarge Retail
2. Acme Corp
3. ENCOM Software
4. Abstergo Industries
5. Wonka Confections

SOURCES

=====

[1] email-wonka-202411251745	2024-11-25	score=0.602
[2] email-buynlarge-202308281000	2023-08-28	score=0.601

Use Case (Project Issues)

```
_ = ask('Why did we credit Acme $4,200? Give me a bullet-point list of reasons')
```

✓ 24.5s

ANSWER

=====

We credited Acme \$4,200 for the following reasons:

- 18 of the 31 contested seats were dormant >60 days at billing snapshot [Source]
- 9 of the 31 contested seats were single-event logins (probably 'check this thing')
- We propose a new billing model in renewal: monthly-active-billed-user rather than

The credit amount (\$4,200) is the math on the 18 dormant seats (~\$2.4k) + half of

No further credits are planned – this covers Acme's billing dispute from Q1-2025

SOURCES

=====

[1] email-acme-003	2025-03-03	score=0.697
[2] email-acme-004	2025-03-03	score=0.697

Use Cases (Understanding Renewal Discussions)

```
_ = ask('List every customer in your knowledge base and their renewal date, sorted by ARR descending.')
```

[10] ✓ 14.1s

... ANSWER

=====

Here are the customers in my knowledge base and their renewal dates:

- Acme Corp: Renewing 2024-11-30 [Source 5]
- BuyNLarge Retail: Renewing 2022-11-30 [Source 6], then every year thereafter
- ENCOM Software: Renewing 2025-09-30 [Source 8]
- Abstergo Industries: Not a named customer, no renewal date found in the provided context. [Source 3]
- Northwind SaaS (our company): Not a named customer, no renewal date found in the provided context. [Source 7]

SOURCES

=====

[1]	email-wonka-202411251745	2024-11-25	score=0.602
[2]	email-buynlarge-202308281000	2023-08-28	score=0.601
[3]	email-abstergo-202505190931	2025-05-19	score=0.595
[4]	email-buynlarge-202408081322	2024-08-08	score=0.594
[5]	email-acme-202407150800	2024-07-15	score=0.593
[6]	email-wonka-202203070938	2022-03-07	score=0.593
[7]	email-buynlarge-202405061300	2024-05-06	score=0.593



Challenges

```
_ = ask('What happened with Hoolie Incident?')
```

✓ 7.9s

ANSWER

=====

Hi Brent,

Thanks for walking through. Let me pull up my side of the room. Will revert with specifics by EOD.

Best,

Priya

P.S. – quick context: I've routed this to my team and our support is aware on the ops side. Sasha (our security lead) has your questionnaire and is on

[Source 1]

SOURCES

=====

[1]	email-hooli-incident-003	2025-05-22	score=0.712
[2]	email-hooli-followup-001	2025-06-04	score=0.654
[3]	email-hooli-202406101745	2024-06-10	score=0.636
[4]	email-hooli-incident-001	2025-05-22	score=0.630
[5]	email-hooli-202308281431	2023-08-28	score=0.630
[6]	email-hooli-202307101400	2023-07-10	score=0.629
[7]	email-hooli-202409100831	2024-09-10	score=0.628

Possible Digital Twins

OPTION 1

One Employee Digital Twin

Personalized AI for each employee



What it does

Captures the employee's voice, style, decisions and knowledge from their documents.

Best for

- Executive assistant
- Sales rep memory
- Personal productivity
- Employee digital twin

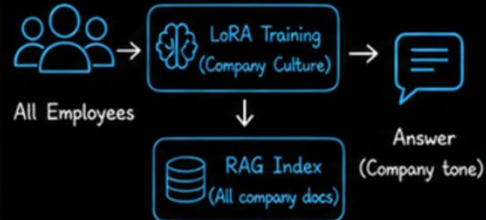
Key Benefit

Highly personalized and realistic.

OPTION 2

One Company Model + Shared RAG

Consistent company assistant for everyone



What it does

One model learns company culture and tone. RAG provides knowledge from all company data.

Best for

- Internal assistant (HR, IT, Policies)
- Onboarding & training
- Support agent
- Company-wide Q&A

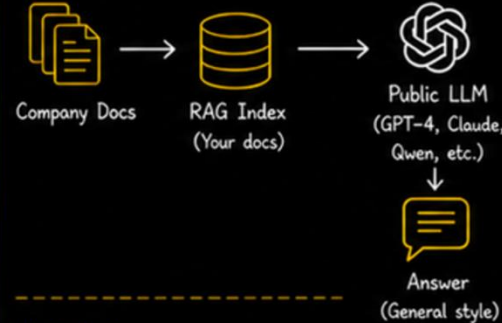
Key Benefit

Scalable, consistent and easy to maintain.

OPTION 3

Public LLM + Private RAG (No Training)

Use powerful public LLM with your data



What it does

No fine-tuning. Just retrieve and ask a strong public model.

Best for

- Quick setup / MVP
- Knowledge search
- FAQ bot
- Cost-effective solutions

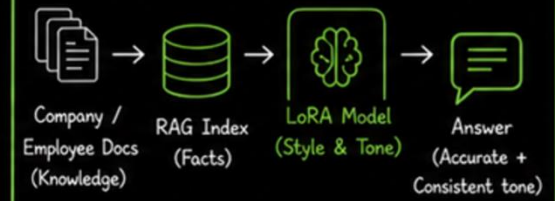
Key Benefit

Fastest and cheapest to deploy.

OPTION 4

Hybrid (Recommended) LoRA (Behavior) + RAG (Knowledge)

Best balance of tone, accuracy & scale



What it does

RAG provides facts, LoRA ensures the right tone, style and behavior.

Best for

- Enterprise AI Assistant
- Customer facing bots
- Employee productivity tools
- Digital twin at scale

Key Benefit

Best balance of accuracy, tone and scalability.

⇒ EmployeeRecall – Conclusion ⇐

① RISKS



- Ethics / Legality / Privacy Risk
- PII exposure of about that employee
- Cross Department Data Exposure
- Compliance / Consent Risk
- Wrong Information can be presented as fact

② WHAT WE DID WITH LORA



- Created a persona AI using LoRA fine-tuning.
- Merged LoRA with the base model and quantized to GGUF.
- Runs locally with Ollama for privacy, speed, and low cost.
- Combined with RAG to answer using only your documents.

③ TRAIN YOUR OWN MODEL



- Prepare your documents (collect & clean).
- Create embeddings & build the RAG index.
- Fine-tune with LoRA on your data.
- Merge LoRA, quantize (GGUF) and run locally.
- Your own persona AI is ready!



Open-source ProjectRecall ⇐

<https://github.com/kader-xai/EmployeeRecall>